Workshop Report: Virtual X-Ray Astronomy data analysis workshop

Naxatra – The astronomy club of IISER Berhampur

July 27-28, 2024

Overview

Naxatra's workshop on X-ray astronomy was held on July 27-28, 2024, attracting 330 undergraduate and post-graduate participants from various institutes across India. It was held as a part of Invenio, the first science fest of IISER Berhampur. This workshop provided a comprehensive platform for students to engage with leading experts in astronomy, focusing on the latest advancements in X-ray astronomy and space-based observational techniques.

Speakers and Affiliations

The workshop featured an impressive lineup of speakers from renowned institutions:

- **Dr. Sreenath Rathnakumar (Main Coordinator)**
  - Scientist SF, ISSDC, ISTRAC ISRO.
- **Dr. Dipankar Bhattacharya**
  - Professor of Astrophysics at Ashoka University.
- **Dr. M.C. Ramadevi**
  - Space Astronomy Group, UR Rao Satellite Centre (URSC), ISRO
- **Dr. Sankarasubramanian K.**
  - Space Astronomy Group, UR Rao Satellite Centre (URSC), ISRO
- **Dr. Radhakrishna V.**
  - Space Astronomy Group, UR Rao Satellite Centre (URSC), ISRO
- **Dr. Shyama Narendranath and Dr. Netra S. Pillai**
  - Space Astronomy Group, UR Rao Satellite Centre (URSC), ISRO

Dr. Vyas Akondi who is our faculty adviser, guided us with every step of the workshop.

Participants

The workshop saw active participation from 330 undergraduate students, from various state and center institutes, who were eager to delve into the realms of astronomy and space science. The gender ratio of the registered participants was close to half.

The workshop sessions were held over zoom virtual meeting platform where the around 100 participants showed up in each session.

The sessions were also live-streamed on our club's non-monetized YouTube channel which also gained many views.

Workshop Schedule

The workshop was meticulously scheduled over two days, with sessions covering various aspects of X-ray astronomy:

- **27th July, 2024**
  - **10:00 – 11:00** *YouTube link*
    Title: Overview of X-ray astronomy (Guest lecture)
    Speaker: Dr. Dipankar Bhattacharya
  - **11:30 – 13:00** *YouTube link*
    Title: X-ray Astronomy using AstroSat
    Speaker: Dr. M.C. Ramadevi
  - **16:00 – 17:30** *YouTube link*
    Title: X-ray Astronomy of the Sun
    Speaker: Dr. Sankarasubramanian K.
- **28th July, 2024**
  - **11:30 – 13:00** *YouTube link*
    Title: X-ray Instruments
    Speaker: Dr. Radhakrishna V.
  - **16:00 – 17:30** *YouTube link*
    Title: Lunar analysis with X-ray astronomy
    Speaker: Dr. Shyama Narendranath and Dr. Netra S Pillai

Topics Covered

The workshop covered a range of topics critical to the field of X-ray astronomy:

- **Overview of X-ray Astronomy:** Dr. Dipankar covered the overall picture of X-ray Astronomy in his guest lecture. He gave a broad description of sources and

detectors, and some generic aspects of space astronomy data while encompassing the global scenario of X-ray astronomy. He motivated the participants about this interesting field.

- **X-ray Astronomy using AstroSat:** Dr. M.C. Ramadevi discussed the capabilities and scientific achievements of AstroSat, India's first dedicated multi-wavelength space observatory. Participants learned about the instrumentation on AstroSat and its role in observing X-ray sources.
- **X-ray Astronomy of the Sun:** Dr. Sankarasubramanian K. presented on the high-energy phenomena of the Sun observed through X-ray astronomy, providing insights into solar flares, coronal mass ejections, and their implications for space weather.
- **X-ray Instruments:** Dr. Radhakrishna V. provided an overview of the technological advancements and innovations in X-ray instruments, highlighting their design, functionality, and applications in space missions.
- **Lunar analysis with X-ray astronomy:** Dr. Shyama Narendranath and Dr. Netra S Pillai discussed the unique perspective offered by lunar platforms for X-ray observations, explaining how the Moon can serve as an ideal location for space telescopes to observe high-energy cosmic phenomena.

Data Analysis

In each session participants were introduced to data analysis techniques used in X-ray astronomy. The workshop included sessions on how to handle and interpret data from space observatories, with a focus on processing X-ray images, spectroscopic data, and time-series analysis. The speakers provided the participants with adequate resources and materials to explore. These sessions equipped students with essential skills for conducting their research and contributing to ongoing projects in astrophysics.

Conclusion

The Invenio Workshop on Astrophysics was a resounding success, providing UG and PG students with a unique opportunity to learn from and interact with leading scientists in the field of X-ray astronomy. The engaging sessions and the high level of participation underscored the enthusiasm and commitment of the young participants towards advancing their knowledge and careers in astronomy. We, as a club were much enthusiastic and thrilled in organizing the workshop and we plan to conduct similar events in the future as this highly reflected our main objective of promoting astronomy and its fields in the community.